

Limit of some special sequence.

Thm. If $p > 0$, $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$

Proof. Let $\epsilon > 0$ be given. Take $N \in \mathbb{N}$ s.t

$$\left(\frac{1}{\epsilon}\right)^p < N \quad \left(\Leftrightarrow \frac{1}{N^p} < \epsilon\right)$$

Then, for all $n \geq N$,

$$\frac{1}{n^p} \leq \frac{1}{N^p} < \epsilon.$$

Thm. If $p > 0$, $\lim_{n \rightarrow \infty} \sqrt[n]{p} = 1$

Proof. If $p = 1$, then trivial.

If $p > 1$, then $x_n = \sqrt[n]{p} - 1 > 0$. By binomial Thm,

$$1 + n^{\frac{p-1}{n}} \leq (1 + x_n)^n = p.$$

Now, $0 < x_n \leq \frac{p-1}{n} \rightarrow 0$ as $n \rightarrow \infty$

Thus, $\lim_{n \rightarrow \infty} \sqrt[n]{p} - 1 = 0$.

If $p < 1$, then,

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{p}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{p}} = 1.$$

Thus $\lim_{n \rightarrow \infty} \sqrt[n]{p} = 1$.

Thm. $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$

Proof. put $x_n = \sqrt[n]{n} - 1$. Then, by ^{the} binomial Thm,

$$n = (x_n + 1)^n \geq \binom{n}{2} x_n^2 = \frac{n(n-1)}{2} \cdot x_n^2$$

Now, $0 < x_n \leq \sqrt{\frac{2n}{n(n-1)}} = \sqrt{\frac{2}{n-1}} \rightarrow 0$ as $n \rightarrow \infty$.

Corollary. For any non-zero $f(x) \in \mathbb{R}[x]$,

$$\lim_{n \rightarrow \infty} \sqrt[n]{f(n)} = 1.$$

Thm. If $p > 0$ and $\alpha \in \mathbb{R}$, then

$$\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1+p)^n} = 0.$$

Proof. Take $k \in \mathbb{Z}$ s.t. $k > \max\{\alpha, 0\}$. Then, for $n \geq 2k$,

$$(1+p)^n > \binom{n}{k} p^k = \frac{n!}{(n-k)! \cdot k!} \cdot p^k > \left(\frac{n}{2}\right)^k \cdot \frac{p^k}{k!}$$

$$\Rightarrow \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{n^k p^k} \cdot n^\alpha = \frac{2^k k!}{p^k} \cdot n^{\alpha-k} \rightarrow 0 \text{ as } n \rightarrow \infty, \text{ since } \alpha - k < 0.$$

Thm. If $0 \leq x < 1$, then

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

Proof. Since

$$(1-x)(1+x+\dots+x^n) = 1-x^{n+1} \Leftrightarrow \sum_{k=1}^n x^k = \frac{1-x^{n+1}}{1-x}$$

And $x^{n+1} \rightarrow 0$ as $n \rightarrow \infty$, we obtain the result.

Thm. Suppose that $\{a_n\}$ is decreasing non-negative number sequence. Then,

$$\sum_{n=1}^{\infty} a_n \text{ converges iff } \sum_{n=1}^{\infty} 2^k a_{2^k} \text{ converges.}$$

Proof. Since $\{a_n\}$ is consisted by non-negative terms, it suffices to find bound for partial sum.

$$\text{Denote } S_n = \sum_{i=1}^n a_i \text{ and } t_k = \sum_{j=1}^k 2^j \cdot a_{2^j}$$

If $n < 2^k$, then

$$\begin{aligned} S_n &= a_1 + a_2 + \dots + a_n \\ &\leq a_1 + a_2 + \dots + a_{2^{k-1}} + a_{2^k} + \dots + a_{2^{k-1}} \\ &\leq a_1 + 2a_2 + 2^2 a_{2^2} + \dots + 2^{k-1} a_{2^{k-1}} \\ &= t_k \end{aligned}$$

If $n \geq 2^k$, then

$$\begin{aligned} S_n &= a_1 + a_2 + \dots + a_n \\ &\geq a_1 + a_2 + \dots + a_{2^{k-1}} + \dots + a_{2^k} \\ &\geq \frac{1}{2} a_1 + a_2 + 2a_{2^2} + 2^2 a_{2^3} + \dots + 2^{k-1} a_{2^k} \\ &= \frac{1}{2} t_k. \end{aligned}$$

Now, if S_n is bounded, then t_k is bounded, similarly converse.

Thm. If $p > 1$, then $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ Converges.

If $p \leq 1$, then diverges.

Proof. Since $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ Converges if and only if

$$\sum_{k=1}^{\infty} 2^k \cdot \frac{1}{2^k (\log 2^k)^p} = \sum_{k=1}^{\infty} \frac{1}{k^p (\log 2)^p} = \frac{1}{(\log 2)^p} \cdot \sum_{k=1}^{\infty} \frac{1}{k^p}$$

we obtain the result.

Thm. For any $c \in \mathbb{R} \setminus \mathbb{Z}$, $\sum_{n=1}^{\infty} \frac{1}{n+c}$ diverges.

Proof. Take $k \in \mathbb{N}$ s.t. $k > c+1$. Then,

$$\sum_{n=1}^{\infty} \frac{1}{n} \leq \sum_{n=k}^{\infty} \frac{1}{n} \leq \sum_{n=1}^{\infty} \frac{1}{n+c}$$

Def. $e \stackrel{\text{def}}{=} \sum_{n=0}^{\infty} \frac{1}{n!}$.

Since $S_n = \sum_{k=0}^n \frac{1}{k!} = 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \dots + \frac{1}{1 \cdot 2 \cdot \dots \cdot n} \leq 1 + \sum_{k=2}^n \frac{1}{2^{k-1}} \rightarrow 3$, well-defined.

Thm. $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$.

Proof. part $S_n = \sum_{k=0}^n \frac{1}{k!}$, $t_n = \left(1 + \frac{1}{n}\right)^n$.

By the binomial theorem,

$$\begin{aligned} t_n &= \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \frac{n!}{(n-k)! \cdot k!} \cdot \frac{1}{n^k} = \sum_{k=0}^n \frac{1}{k!} \cdot \frac{n \cdot (n-1) \cdot \dots \cdot (n-k+1)}{n^k} \\ &= \sum_{k=0}^n \frac{1}{k!} \cdot \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{2}{n}\right) \cdot \dots \cdot \left(1 - \frac{k-1}{n}\right) \\ &\leq \sum_{k=0}^n \frac{1}{k!} = S_n. \end{aligned}$$

$$n \geq m$$

$$\frac{1}{n} \leq \frac{1}{m}$$

So that, $\limsup_{n \rightarrow \infty} t_n \leq \limsup_{n \rightarrow \infty} S_n = \lim_{m \rightarrow \infty} S_m = e$.

$$\left(1 - \frac{1}{n}\right) \geq \left(1 - \frac{1}{m}\right)$$

Meanwhile, if $n \geq m$, then

$$\begin{aligned} t_n &= \sum_{k=0}^n \frac{1}{k!} \cdot \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{2}{n}\right) \cdot \dots \cdot \left(1 - \frac{k-1}{n}\right) \\ &\geq \sum_{k=0}^m \frac{1}{k!} \cdot \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{2}{n}\right) \cdot \dots \cdot \left(1 - \frac{k-1}{n}\right) \quad (\text{Note: this is not } t_m) \\ &\geq \sum_{k=0}^m \frac{1}{k!} = S_m \end{aligned}$$

Therefore, for any $m \in \mathbb{N}$, $\liminf_{n \rightarrow \infty} t_n \geq S_m$

Now, $\liminf_{n \rightarrow \infty} t_n \geq \lim_{m \rightarrow \infty} \inf_{n \rightarrow \infty} S_m = \lim_{m \rightarrow \infty} S_m = e$.

Consequently,

$$e \leq \liminf_{n \rightarrow \infty} t_n \leq \limsup_{n \rightarrow \infty} t_n \leq e.$$

Note: for all $n \in \mathbb{N}$,

$$e - S_n = \sum_{k=n+1}^{\infty} \frac{1}{k!} = \frac{1}{(n+1)!} \cdot \sum_{k=0}^{\infty} \frac{1}{k!} < \frac{1}{(n+1)!} \sum_{k=0}^{\infty} \frac{1}{(1+k)^k} = \frac{1}{n! \cdot n}$$

Thm. e is irrational.

Proof. Suppose that e is rational. Since e is positive, $e = \frac{p}{q}$ for some $p, q \in \mathbb{N}$.

Denote $S_n = \sum_{k=0}^n \frac{1}{k!}$. Then, by approximation of e ,

$$0 < q!(e - S_n) < \frac{1}{q}.$$

By assumption, $q!e = p \cdot (q-1)!$ is an integer, and

$$q! \cdot S_n = \sum_{k=0}^n \frac{q!}{k!}$$

is integer, being sum of integers,

Now, $q!(e - S_n)$ is an integer between 0 and 1, it's contradiction.

Prop. Let $S_1 = \sqrt{2}$, and $S_{n+1} = \sqrt{2 + \sqrt{S_n}}$ ($n \geq 1$).

Then, S_n converges.

Proof. At first, $\sqrt{2} \leq S_1 = \sqrt{2} < 2$.

Suppose that for $n \in \mathbb{N}$, $\sqrt{2} < S_n < 2$. Then,

$$\sqrt{2} < S_{n+1} = \sqrt{2 + \sqrt{S_n}} < \sqrt{2 + \sqrt{4}} = 2$$

Thus, $\{S_n\}$ lies in $(\sqrt{2}, 2)$.

$$\text{And, } S_2 = \sqrt{2 + \sqrt{S_1}} = \sqrt{2 + \sqrt{\sqrt{2}}} > \sqrt{2} = S_1.$$

Suppose that for $n \geq 2$, $S_n > S_{n-1}$.

$$\text{Then, } S_{n+1} = \sqrt{2 + \sqrt{S_n}} > \sqrt{2 + \sqrt{S_{n-1}}} = S_n$$

Now, $\{S_n\}$ is increasing bounded sequence, thus converges.

Prop. If $\{a_n\}$ is non-negative and $\sum a_n$ converges, then $\sum \frac{\sqrt{a_n}}{n}$ converges.

Proof. observe that: for all $n \in \mathbb{N}$,

$$0 \leq (n\sqrt{a_n} - 1)^2 = n^2 a_n - 2n\sqrt{a_n} + 1 \Leftrightarrow \frac{\sqrt{a_n}}{n} \leq \frac{1}{2} \cdot \left(a_n + \frac{1}{n^2}\right)$$

or using Cauchy-Schwarz inequality:

$$\sum_{n=1}^N \frac{\sqrt{a_n}}{n} \leq \sqrt{\sum_{n=1}^N a_n} \cdot \sqrt{\sum_{n=1}^N \frac{1}{n^2}}.$$

Lemma. Cauchy-Schwarz inequality

$$\sum_{i=1}^n |a_i b_i| \leq \sqrt{\sum_{i=1}^n a_i^2} \cdot \sqrt{\sum_{i=1}^n b_i^2}$$

Prop. Suppose $\sum a_n$ converges absolutely.

Then, the following series are absolutely convergent:

$$\sum_{n=1}^{\infty} a_n^2, \quad \sum_{n=1}^{\infty} \frac{a_n}{1+a_n} \quad (a_n \neq -1), \quad \sum_{n=1}^{\infty} \frac{a_n^2}{1+a_n^2}$$

Proof. Since $\sum a_n$ converges absolutely, there exists $N \in \mathbb{N}$ s.t.

$$n \geq N_1 \Rightarrow |a_n| < \frac{1}{2}$$

Now since $x^2 \leq x$ for $0 \leq x \leq 1$, $n \geq N_1$ implies

$$\sum_{n=1}^m |a_n^2| \leq \sum_{n=1}^{N_1} |a_n^2| + \sum_{n=N_1}^m |a_n^2| \leq \sum_{n=1}^{N_1} |a_n^2| + \sum_{n=N_1}^m |a_n| \leq \sum_{n=1}^{N_1} |a_n^2| + \sum_{n=N_1}^{\infty} |a_n|$$

This implies that $\sum \frac{a_n^2}{1+a_n^2}$ converges directly:

$$\sum_{n=1}^{\infty} \left| \frac{a_n^2}{1+a_n^2} \right| = \sum_{n=1}^{\infty} \frac{a_n^2}{1+a_n^2} \leq \sum_{n=1}^{\infty} a_n^2.$$

Meanwhile, $|a_n| < \frac{1}{2}$ implies

$$\frac{1}{2} < |1+a_n| < \frac{3}{2} \Leftrightarrow \frac{2}{3} < \frac{1}{|1+a_n|} < 2$$

That is, for $n \geq N_1$, $\left| \frac{a_n}{1+a_n} \right| \leq 2|a_n|$, thus $\sum_{n=1}^{\infty} \frac{a_n}{1+a_n}$ converges absolutely.

Prop. If $\sum a_n$ converges and $\{b_n\}$ is monotonic and bounded,
then $\sum a_n b_n$ converges.

Proof. Since $\{b_n\}$ is monotonic and bounded, it is convergent. Let $b = \lim_{n \rightarrow \infty} b_n$.

Suppose that $\{b_n\}$ is decreasing. Then, for all $n \in \mathbb{N}$, $b_n \geq b$.

Now, the sequence $\{b_n - b\}$ is monotonically decreasing and $b_n - b \rightarrow 0$ as $n \rightarrow \infty$.

Therefore, $\sum a_n(b_n - b)$ converges.

Counterexample if $\{b_n\}$ converges but not monotone.

$\sum (-1)^n \frac{1}{\sqrt{n}}$ converges, and $b_n = (-1)^n \frac{1}{\sqrt{n}}$ converges.

But $\sum \frac{1}{n}$ not converges.

